

Pankaj Kumar

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About

Backend Engineer with 5+ years of experience building scalable, high-performance, and data-intensive backend systems. Experienced in designing and developing services that support **millions** of users, with strong expertise in **concurrency**, **multi-threading**, and **distributed system design**, with a focus on building efficient, reliable, and scalable software systems. I am good at solving problems in a clear and organized way, and possess a strong foundation in Data Structures & Algorithms and System Design.

Education

IIT Kanpur, Dual Degree in Mathematics and Computer Science

July 2015 – May 2020

- **CPI:** 9.3/10
- **Coursework:** Introduction to Programming, Data structures and algorithms
- **Online courses** Introduction to GoLang, Java Multithreading, Java Advanced Concepts, Java Memory Management, Redis Data Structures, Apache Kafka Fundamentals, Fundamentals of Back-end Engineering

Experience

Software Engineer II, Arista Networks – Bangalore, India

Mar 2024 – Present

- **Campus-AGNI Integration**
 - Designed and developed the **Campus App** to process large-scale data and handle real-time event subscriptions, supporting over **9–10K active user-device sessions daily** across on-prem and multi-tenant deployments, processing **~1K events/sec**.
 - Integrated with a third-party system (**AGNI**) to fetch and synchronize users, user groups, clients (IoT/user devices), client groups, and session data (Auth/IP/Disconnect) using **Go** and an **HBase** database, enabling synchronization of **10K+ records** with reliable near real-time updates.
 - Conducted a **POC in Python** to predict and calculate in-memory cache size, preventing application restarts caused by insufficient memory and improving system stability by reducing memory-related failures by **60%**.
 - Developed a highly flexible **AGNI Simulator** to support integration and stress testing for end-to-end product and load validation, capable of simulating **thousands of concurrent clients and sessions**.
 - Added multiple **ptests** and **ctests** in Python to automate testing and enable real-device setups for product validation, improving **test coverage by 40%+** and reducing manual validation efforts.
 - Owned the complete lifecycle — **design, development, testing** (ptest, ctest, stest, unit, integration, smoke), and **deployment** — ensuring reliable releases and reducing post-deployment issues.
 - Added **observability and monitoring** using **Prometheus** and **Grafana**, with alerting configured for critical metrics (latency, in-memory size, event backlog, session failures), enabling faster detection and resolution of production issues.
- **Asset Management App**
 - Designed and implemented an **Asset Management App** to manage purchased devices globally across multiple tenants and users within the same organization.
 - Built for a large enterprise managing around **8–10K devices** in a multi-tenant setup.
 - Took full ownership of **design, development, testing** (unit, integration, smoke), and **deployment**.
- **License Management App**
 - Built and optimized a **License Management App** that automatically installs and manages licenses for over **300K devices**, reducing manual intervention in multi-tenant deployments.
 - Responsible for **development, code review, and testing** (unit, integration, smoke).
- **Tech Stack:** Go, Python, Java | Springboot, Flask | Kafka | HBase, Redis | Grafana | Prometheus | AWS

Software Engineer II, Ocrolus – Gurgaon, India

Mar 2023 – Feb 2024

- Developed a **Fraud Detection Application** that identifies suspicious documents, processing over **50K documents daily**. Built using **Python (Flask)** with a **PostgreSQL** backend and comprehensive unit tests implemented via **pytest**.
- Designed a **generic, scalable workflow** leveraging **SOLID principles** and **OOP concepts** to generate fraud detection signals and visualizations across multiple document types including paystubs, driver's licenses, W2 forms, and bank statements.
- Implemented automated maintenance using a **Celery-based cron job** to regularly remove unverified and unsupported documents, preventing database misuse and ensuring data integrity.
- **Tech Stack:** Python | Flask | Pytest | Kafka, SQS | PostgreSQL, Redis | Grafana | AWS

Software Engineer I, Cleartax – Bangalore, India

Aug 2020 – Feb 2023

- **Indirect Tax Filing Platform Rewrite**
 - Built an end-to-end **Indirect Tax Filing Platform** from scratch, handling over **100K transactions per month**. The system was modularized into three key components: (1) data upload to the ClearTax platform, (2) data submission and GST filing, and (3) reconciliation using GSTR 2A and 2B for data verification.
 - Integrated multiple **Government APIs** for GST data upload, retrieval, and submission, covering forms such as **G1, G3B, G6A, G7A, G4B, G9, G9C, 2A, 2B, TP, and RETTRACK**.
 - Implemented **unit and integration tests**, set up automated testing and **CI/CD pipelines**, and configured monitoring tools for **metrics, alerts, logs, and dashboards**.
- **Licensing and Metering Service**
 - Designed and developed a **Licensing and Metering Service** to track and enforce license usage, reducing unauthorized access by **40%**.
 - Responsible for designing, developing, testing (unit test, integration test), and deploying product code with full ownership
- Refactored legacy code by applying **Design Patterns** and **SOLID principles**, reducing API latency by **25%** through optimized queries, indexed tables, and removal of redundant DB calls.
- Improved the on-call process by conducting training sessions, providing better tooling, and automating tasks, resulting in a **30%** reduction in ticket volume.
- **Tech Stack:** Java (DropWizard, Spring Boot) | Maven | Mockito, JUnit | JPA, Hibernate | Kafka, SQS | Jenkins, Docker, Kubernetes | MySQL, MongoDB, InfluxDB, Redis | CI/CD, Git | AWS (S3, EC2, Route 53) | Grafana, Coralogix, Scalyr, New Relic, Metabase, Retool, SonarQube, Vault, Liquibase

Skills

Languages: Golang, Java, Python, C/C++

Databases: MySQL, Postgres, MongoDB, H-base, Redis, InfluxDB

Message Queues: Kafka, SQS

AWS: Route 53, ECR, EC2, S3, Lambda, CloudWatch

Deployment: CI/CD, Jenkins, Docker, Kubernetes, Git

Tools: Maven, Bazel, Junit, Mockito, Pytest, JPA, Hibernate, SQLAlchemy, REST

Achievement

Hackathon Winner: Developed an LLM-powered model to be the first PR reviewer at **Arista Networks**

Spot Award Winner: Recognized for outstanding contributions to the license service project at **Cleartax**